APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination

MODEL QUESTION PAPER — SET 4

Course Code: PCCST501 **Course Name:** Computer Networks

Maximum Marks: 60 **Duration:** 2 hours 30 minutes

PART A — Answer all questions. Each question carries 3 marks.

Q. No.	Question	Marks	CO
1	Explain the purpose of the peer-to-peer paradigm in application-layer communication.	3	CO1
2	Describe the role of electronic mail as an application-layer service.	3	CO1
3	Differentiate between TCP connection establishment and UDP datagram transmission.	3	CO2
4	Explain how the poll() function helps a server monitor multiple file descriptors.	3	CO2
5	Explain the role of multicast routing in delivering packets to a group of hosts.	3	CO3
6	Describe how a routing cache differs from a routing table in packet forwarding.	3	CO3
7	Explain the purpose of connecting devices in a data-link network.	3	CO4
8	Distinguish between analog and digital signals.	3	CO5

PART B — Answer one full question from each module. Each question carries 9 marks.

Module 1 — Question 9 (CO: CO1; Marks: 9)

(a) Explain the role of electronic mail and the function of DNS in supporting application-layer communication. (4 marks)
(b) A user sends an email to a recipient using a domain name. Describe the roles of the mail application protocol and DNS in locating the destination mail service. (5 marks)

Module 1 — Question 10 (CO: CO1; Marks: 9)

(a) Compare client-server and peer-to-peer paradigms in terms of resource location and responsibility for service provision. (4 marks)
(b) Explain how BitTorrent enables peers to exchange file pieces without relying on one host to supply the entire file to every user. (5 marks)

Module 2 — Question 11 (CO: CO2; Marks: 9)

(a) Explain the purpose of TCP connection establishment and identify the information exchanged during the three-way handshake. (5 marks)
(b) A TCP sender begins with sequence number 5000 and sends two segments carrying 300 bytes each in order. Determine the sequence number of the first byte in the third segment. (4 marks)

Module 2 — Question 12 (CO: CO3; Marks: 9)

(a) Explain how routing protocols support the construction of forwarding tables for unicast routing. (4 marks)
(b) A network has these link costs: R–S = 2, R–T = 6, S–T = 1, S–U = 5, T–U = 2. Compute the least-cost path from R to U and its total cost. (5 marks)

Module 3 — Question 13 (CO: CO4; Marks: 9)

(a) Explain the role of multiple-access protocols in sharing a communication channel among several devices. (5 marks)
(b) Describe how a data-link frame uses link-layer addresses to support delivery on an Ethernet LAN. (4 marks)

Module 3 — Question 14 (CO: CO4; Marks: 9)

(a) Explain the functions of SOCK_PACKET and PF_PACKET in packet-oriented data-link access as named in the syllabus. (5 marks)
(b) Describe the purpose of the Data-Link Provider Interface in accessing data-link services. (4 marks)

Module 4 — Question 15 (CO: CO5; Marks: 9)

(a) Explain how digital transmission represents data using signals and identify two factors that affect its quality. (5 marks)
(b) A link operates at 5,000 bits per second. How many bits can it transmit in 12 seconds, assuming the rate remains constant? (4 marks)

Module 4 — Question 16 (CO: CO5; Marks: 9)

(a) Explain the relationship between bandwidth utilization, transmission media and the quality of received signals. (5 marks)
(b) A signal has a frequency of 2.5 kHz. Express this frequency in hertz and state what frequency measures. (4 marks)

— End of Question Paper —

B. BLUEPRINT TABLE

Q. No.	Module	Syllabus topic	CO	Bloom's level	Marks
1	1	Peer-to-peer application paradigm	CO1	Understand	3
2	1	Electronic mail	CO1	Understand	3
3	2	TCP and UDP	CO2	Understand	3
4	2	I/O multiplexing: poll()	CO2	Understand	3
5	2	Multicast routing	CO3	Understand	3
6	2	Routing table and routing cache	CO3	Understand	3
7	3	Connecting devices	CO4	Understand	3
8	4	Data and signals	CO5	Understand	3
9	1	Electronic mail; DNS	CO1	Apply	9
10	1	Client-server, P2P, BitTorrent	CO1	Analyse	9
11	2	TCP connection establishment; sequence numbers	CO2	Apply	9
12	2	Unicast routing; forwarding tables	CO3	Apply	9
13	3	Multiple-access protocols; Ethernet addressing	CO4	Apply	9
14	3	SOCK_PACKET, PF_PACKET; Data-Link Provider Interface	CO4	Understand/Analyse	9
15	4	Digital transmission; data rate	CO5	Apply	9
16	4	Bandwidth utilization; transmission media; frequency	CO5	Apply	9

C. VERIFICATION CHECKLIST

Check	Result
Part A marks	8 × 3 = 24 marks.
Part B marks	Four answered questions × 9 = 36 marks.
Module distribution	Exactly two Part A questions and two Part B alternatives per module.
Subdivision totals	Q9–Q16 each total 9 marks and contain no more than two subdivisions.
CO coverage	CO1, CO2, CO3, CO4 and CO5 are assessed.
Problem-solving	Q11 sequence-number calculation, Q12 least-cost route, Q15 data-volume calculation.
Numerical verification	Q11: third segment starts at 5000 + 2×300 = 5600. Q12: R–S–T–U costs 2+1+2 = 5. Q15: 5000×12 = 60,000 bits. Q16: 2.5 kHz = 2500 Hz.
No repetition	Questions and scenarios are newly composed for Set 4, with no direct repeats of Sets 1–3.

Official Course Outcomes

CO1: Understand the internetworking design in terms of protocol stack and the role of various application layer protocols

CO2: Illustrate the functions of the transport layer from connectionless and connection-oriented perspectives

CO3: Identify how the network layer achieves host-to-host connectivity and caters to the diverse service requirements of the host applications

CO4: Explain the nuances of the data link layer design and demonstrate the various data link link layer protocols

CO5: Describe the fundamental characteristics of the physical layer and understand how the physical layer supports the functionalities of the top layers

D. SCHEME OF VALUATION

Part A

Q1: P2P distributes participation among peers (1); peers can both request and provide resources (1); valid benefit such as distributed load (1).

Q2: Email supports message exchange between users (1); identifies application-layer mail service/protocol role (1); valid example or explanation (1).

Q3: TCP establishes connection/state before data transfer (1.5); UDP sends independent datagrams without connection setup (1.5).

Q4: poll() receives an array of descriptors/events (1); reports readiness (1); lets server handle multiple descriptors (1).

Q5: Multicast targets a group (1); avoids separate independent unicast delivery to every member where multicast is supported (1); identifies group-routing purpose (1).

Q6: Routing table stores destination/next-hop forwarding information (1.5); routing cache stores reusable route lookup results (1.5).

Q7: Connecting devices interconnect hosts/links (1); forward or relay data (1); enable local network connectivity (1).

Q8: Analog signal varies continuously (1); digital signal uses discrete levels (1); valid difference in representation/noise handling (1).

Part B

Q9(a) — 4: Email supports message exchange (2); DNS locates domain-related host/service information (2).

Q9(b) — 5: Mail application submits message (1); DNS resolves destination domain/mail-server information (2); mail protocol communicates with destination mail service (2).

Q10(a) — 4: Client-server uses centralized service provider(s) (2); P2P distributes service/resource provision among peers (2).

Q10(b) — 5: File split into pieces (1); peers obtain pieces from other peers (2); peers upload pieces to others (1); pieces combine into complete file (1).

Q11(a) — 5: SYN initiates (1); server acknowledges and sends SYN (2); client acknowledges server SYN (1); establishes connection state/initial sequence coordination (1).

Q11(b) — 4: Each segment carries 300 bytes (1); two segments carry 600 bytes (1); 5000+600 calculation (1); answer 5600 (1).

Q12(a) — 4: Routing protocols exchange/derive reachability information (2); forwarding table maps destinations to next hops/interfaces (2).

Q12(b) — 5: Path R–S–T–U (1); cost 2+1+2 (2); total 5 (1); compares other plausible paths and confirms minimum (1).

Q13(a) — 5: Shared-channel contention problem (1); protocol coordinates access (1); describes valid access mechanism (2); efficiency/collision handling (1).

Q13(b) — 4: Source and destination link-layer addresses (2); frame encapsulates payload (1); addresses support local frame delivery (1).

Q14(a) — 5: Identifies packet-socket purpose (1); PF_PACKET packet-family interface (2); SOCK_PACKET older interface (1); valid distinction in use/role (1).

Q14(b) — 4: DLPI provides interface to data-link services (2); supports service access/control (1); relationship to link-layer protocol operation (1).

Q15(a) — 5: Digital data mapped to discrete signal states (2); identifies two valid quality factors such as noise, attenuation or bandwidth (2); explains effect (1).

Q15(b) — 4: Bits = rate×time (1); substitution 5000×12 (2); answer 60,000 bits (1).

Q16(a) — 5: Bandwidth relates to available frequency range/capacity (2); medium affects attenuation/noise (2); signal-quality consequence (1).

Q16(b) — 4: 1 kHz = 1000 Hz (2); 2.5 kHz = 2500 Hz (1); frequency measures cycles per second (1).